

FFGFT in Hilbert Space

From the Shared Loop Object to the Mass Operator

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- $1/\alpha$: "corpus tunes $E_0 = 7.398$, near one percent radiative correction" $\rightarrow 7.398 = 7.348/\sqrt{K_{\text{frak}}}$ (R72), gap $1.35\% \approx 100\xi = 1.33\%$.
- 1 Oct. 2026: numerical caution, m_μ/m_e with rounded cosine $63 \rightarrow \approx 70$; position of $\theta = 2/9$ rad "2/3 between 0 and $1/3$ " $\rightarrow$ about 21 % between 0 and $\pi/3$; the fixed-point condition $\xi_0^{p_i-1} K_{\text{frak}} = 1$ does not determine ξ_0 (previously "fixes ξ_0 ").
- (2 Oct. 2026) Notation f : mean $\langle f \rangle$ of the correction function — not a frequency; note in the text (Doc. 190, R147).

Abstract

The existing Hilbert-space bridge of FFGFT (Doc. 230/231/232) already maps previously described QM operations – but only those, and without time. This document makes the translation more complete and shows that it reaches the *masses*. Three levels: (0) QM-equivalent – both FFGFT and HLV are a connection Laplacian on a fibre-valued Hilbert space (gauge freedom = unitary equivalence, entanglement = holonomy around cycles). (1) Process – a CP-divisibility test computably separates QM-internal (Markovian) from history-coupled (non-Markovian); this is time as process structure. (2) Mass – the actual new finding: the mass operator is a Hermitian Z_3 -circulant on the $\mathbb{C}^3$ fibre, its eigenvectors are the three Z_3 characters (the generations), its spectrum S^2 are the masses. With two pure numbers – $r = \sqrt{2}$ (Koide $Q = 2/3$) and the angle $\theta = 2/9$ – it reproduces m_μ/m_e and m_τ/m_μ to four–five digits, without the formerly fitted ladder coefficient. Here $2/9 = 2/3^2$ is not a tuned decimal but a structural number on the footing of ξ : a foundational condition of the fractal open space. The FFGFT $\leftrightarrow$ HLV comparison is only the entry door; no theoretical equivalence is claimed.

.1 Occasion and scope

The exchange with Marcel Krüger (HLV/HICE) raised the question whether FFGFT and HLV are connected. Pursuing it surfaced the essential point: FFGFT's Hilbert-space mapping is *incomplete* – it translates the already-described QM operations, but *time* does not appear in it, although it very much exists in the real world and in the T^4 tori. And as long as time is missing, the *masses* are missing. This document closes that gap as far as is honestly possible at present.

Scope rule

No theoretical equivalence FFGFT = HLV is claimed. The HLV comparison only supplies the shared object (connection Laplacian) and the process observable (non-Markovianity). The topic is the completion of FFGFT's own Hilbert-space translation up to the masses.

.2 The Hilbert space explicitly

Before computing, the space itself. FFGFT's full Hilbert space is the tensor product of base and fibre:

$$\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3. \quad (1)$$

Bases and inner product

Base $L^2(T^4)$ – functions on the four-torus, two natural bases:

- node basis $|x\rangle$, $x \in T^4$ (nodes of the discrete torus graph);
- winding basis $|n\rangle$, $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$, since $H_1(T^4) = \mathbb{Z}^4$ (Doc. 189) – eigenstates of the translation generators, the n_i being the winding numbers.

Fibre $\mathbb{C}^3$ – the three Z_3 -symmetric ICE channels (Doc. 206), two bases:

- channel basis $|c\rangle$, $c = 1, 2, 3$;
- Z_3 character basis $|f_k\rangle = \frac{1}{\sqrt{3}} \sum_c \omega^{kc} |c\rangle$, $k = 0, 1, 2$, $\omega = e^{2\pi i/3}$ – the *generations*.

Inner product: $\langle \psi | \phi \rangle = \sum_{x \in T^4} \sum_c \overline{\psi(x, c)} \phi(x, c)$. A general state: $|\psi\rangle = \sum_{n,k} a_{n,k} |n\rangle \otimes |f_k\rangle$.

Three operator families act on this space – and they split the physics cleanly:

The three operators on $\mathcal{H}$

(1) Connection Laplacian L on the base (gauge/holonomy sector): $L = D - \sum_{\langle vw \rangle} U_{vw}$, $U_{vw} = e^{iA_{vw}}$. Eigenvalues = internal mode numbers; gauge transformation = unitary conjugation; holonomy around cycles = entanglement (Doc. 230). Lives on $L^2(T^4)$.

(2) Time generator H_t (energy/mass sector): generator of translation in the $\tilde{T}$ direction of the torus. Via $\tilde{T} \cdot m = 1$ its eigenvalue is the rest mass; it sets the scale $M = 1/\tilde{T}$.

(3) Mass operator S on the fibre (generation sector): the Hermitian Z_3 -circulant on $\mathbb{C}^3$. Eigenvalues $\sqrt{m_k}$, eigenvectors = Z_3 characters = generations.

Thus the physics splits: the base $L^2(T^4)$ carries gauge structure, entanglement and dynamics; the fibre $\mathbb{C}^3$ carries generation and mass; the $\tilde{T}$ direction sets the scale. The full operator of a sector is a product (base operator) $\otimes$ (fibre Z_3 -circulant).

.3 Level 0 – the shared object: connection Laplacian

At its core the QM mapping is a connection Laplacian on a fibre-valued function space over a discrete base – the T^4 for FFGFT, the cell graph for HLV:

$$L = D - \sum_{\langle vw \rangle} U_{vw}, \quad U_{vw} = e^{iA_{vw}} \in \mathbf{U}(n), \quad \Omega(C) = \sum_{e \in C} A_e \pmod{2\pi}. \quad (2)$$

What FFGFT achieves at this level

States = fibre-valued fields; QM operation = unitary parallel transport U_{vw} ; gauge freedom = unitary equivalence (gauge-null, passed on both sides, $\sim 10^{-14}$); entanglement = holonomy around cycles (Doc. 230) with $H_1(T^4) = \mathbb{Z}^4$ (Doc. 189); CP phase = accumulated torsion phase (Doc. 172). Counterparts: FFGFT loop Hamiltonian (Doc. 231/232) $\leftrightarrow$ HLV L_{HICE} .

This is solid, but it is only QM cast as a lattice gauge model – and it contains neither time nor mass.

.4 Level 1 – the process level: CP-divisibility

The first extension beyond static QM is time as *process*. A fibre qubit is dephased by a bath whose frequencies $\omega_k = \sqrt{\lambda_k}$ are the loop frequencies of the T^4 connection Laplacian (independent boson):

$$\Gamma(t) = \sum_k \frac{2g_k^2}{\omega_k^2} (1 - \cos \omega_k t), \quad c(t) = e^{-\Gamma(t)}, \quad \gamma(t) = \dot{\Gamma}(t) = \sum_k \frac{2g_k^2}{\omega_k} \sin \omega_k t. \quad (3)$$

Two reduced descriptions of the same operator are compared: (M) memoryless / time-local (constant rate $\Rightarrow$ CP-divisible) and (NM) history-coupled (full T^4 bath $\Rightarrow$ may violate CP-divisibility). Two standard witnesses (BLP trace-distance back-flow, RHP/Choi):

Description	BLP back-flow	D revives	min Choi eig	$\gamma < 0$
(M) memoryless / time-local	0.000	no	+0.013	—
(NM) history-coupled (T^4)	5.125	yes	-0.041	1928

Only the history-coupled variant leaves the Markovian class. This is the operational “beyond-Markov” signature – but beyond the time-local *reduced* description, not beyond quantum theory.

.5 Level 2 – time and mass: the Hilbert-space mass operator

Here lies the actual finding. A mass is a rest energy: the eigenvalue of the *time-translation generator* (Casimir $m^2 = E^2 - p^2$). The basic relation $\tilde{T} \cdot m = 1$ makes mass the *dual* of the time cycle. A static graph Laplacian has discarded time – its eigenvalues are internal mode numbers, not masses. Bringing time back yields masses; and since $\hbar, c$ are merely SI conversion factors, the physical content is the *ratio*, while the scale $M = 1/\tilde{T}$ is set by the time cycle.

A plain Laplacian is not enough

A symmetric T^4 connection Laplacian with $m^2 = \lambda_k$ gives the three lowest “masses” in the ratio 1:1:1.46 and Koide $Q = 0.345 - O(1)$ ratios, no hierarchy, Koide far from $2/3$. Time as a uniform extra dimension does not suffice; the Z_3 channel structure is required.

The operator

The fibre is $\mathbb{C}^3$ – the three Z_3 -symmetric ICE channels (Doc. 206: $\det A_\xi = 1 - \xi$), i.e. the generation space. The full space is $L^2(T^4) \otimes \mathbb{C}^3$; the $\tilde{T}$ cycle of the torus sets the scale. The mass operator is a *Hermitian Z_3 -circulant*

$$S = \text{circ}(t_0, t_1, t_2), \quad t_0 = M \in \mathbb{R}, \quad t_2 = \bar{t}_1, \quad (4)$$

with eigenvalues

$$\mu_k = t_0 + 2|t_1| \cos(\arg t_1 + \frac{2\pi k}{3}) = M(1 + r \cos(\theta + \frac{2\pi k}{3})), \quad k = 0, 1, 2, \quad (5)$$

where $r = 2|t_1|/M$ and $\theta = \arg t_1$. The *eigenvectors* are the three Z_3 Fourier characters $f_k = (1, \omega^k, \omega^{2k})/\sqrt{3}$, $\omega = e^{2\pi i/3}$ – numerically verified with overlap 1.0000. The three generations are the Z_3 characters; the masses are the spectrum of S^2 .

Koide from the operator

With $\sqrt{m_k} = \mu_k$ the Koide identity follows directly:

$$Q = \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} = \frac{1 + r^2/2}{3} \implies Q = \frac{2}{3} \iff r = \sqrt{2}. \quad (6)$$

Koide $Q = 2/3$ is thus nothing but the operator condition $r = \sqrt{2}$ – the $Z_3 + \sqrt{2}$ structure (Doc. 258/259), not a fit.

The lepton masses from two pure numbers

With $r = \sqrt{2}$ and $\theta = 2/9$ – two pure numbers, no fitted coefficient:

	operator	data	error
m_μ/m_e	206.770	206.768	0.00 %
m_τ/m_μ	16.818	16.817	0.01 %
Koide Q	0.666667	0.666661	exactly $2/3$

More than a fit

The formerly fitted ladder coefficient ($r_e \approx 1.349$, Doc. 006) has *vanished* – replaced by the structural number $2/9$. And it is more than an adjustment: *one* angle θ reproduces *both* independent ratios simultaneously (other θ break m_μ/m_e at once). Two pure numbers $\rightarrow$ three masses to 4–5 digits.

Numerical caution: the electron is a cancellation

The lightest eigenvalue is a near-total cancellation, $\mu_e = M(1 + r \cos(\theta + \frac{2\pi}{3})) = M(1 - 0.95965) = 0.04035 M$ – the difference of two nearly equal numbers. Rounding the cosines to only three digits shifts μ_e by orders of magnitude: taking $\cos(\theta + \frac{2\pi}{3})$ as -0.658 instead of the correct -0.6786 (a 3 % error) turns $\mu_e = 0.040$ into 0.069 (a 72 % error), and since $m_e = \mu_e^2$ this flips m_μ/m_e from 207 to ≈ 70 . (*Corrected on 1 Oct. 2026: previously “to 63”* – $\mu_e = 1 + \sqrt{2} \cdot (-0.658) = 0.0694$, $\mu_\mu = 0.5802$, hence $m_\mu/m_e = (0.5802/0.0694)^2 = 69.8$.) The ratios must therefore be checked at *full machine precision*; precisely because the electron is the lightest state, m_μ/m_e is the sharpest test of whether θ is exactly $2/9$ (script `ffgft_mass_operator_C3.py`, numpy precision).

How precise? Against the high-precision e/μ data the angle is confirmed to $|\theta - 2/9| \approx 1.8 \cdot 10^{-7}$ – seven significant figures. The lepton masses are among the most precisely measured quantities in physics; the data will not get sharper. The empirical question is therefore settled: $2/9$ is the number.

$2/9$ as a foundational condition

Must $2/9$ be derived from something deeper? No. The decisive distinction is between a *fitted continuous knob* (bad – that was r_e) and a *fixed structural rational* forced by the geometry (fine – like $2/3$, like $2\pi/3$, like the three-channel count itself).

Structural number, not a fit

$2/9 = 2/3^2$ is a clean rational, not a tuned decimal – exactly the kind of primitive the fractal open space can carry, *on the footing of ξ* . Like $\xi = 4/30000$ it needs no deeper derivation; it is a foundational condition of the geometry.

This shifts the leptons from a ξ -power story to a Z_3 geometry story: the ratios hinge on the pure numbers $\sqrt{2}$ and $2/9$, not on ξ (which does not appear in the ratio). This fits the fact that the ξ -ladder was the weak part for leptons anyway. The fractal space then carries a small set of primitives: ξ for the coupling sector, $(\sqrt{2}, 2/9)$ for the generation sector.

.6 Computing in the Hilbert space: recipe and examples

The procedure is always the same – every physical quantity is an eigenvalue, an eigenvalue relation, or an overlap of eigenbases on $\mathcal{H}$.

The recipe – four steps

1. **Choose the sector** – which quantity (masses, a process observable, a mixing angle, ...).
2. **Write the operator** – the corresponding self-adjoint operator on $\mathcal{H}$ (Z_3 -circulant on $\mathbb{C}^3$ for masses; time-extended connection Laplacian for dynamics; overlap of two eigenbases for mixing).
3. **Diagonalise** – eigenvalues = quantities (as dimensionless ratios), eigenvectors = states; relations (Koide) follow algebraically.
4. **Scale + SI** – scale $M = 1/\tilde{T}$ from the time cycle, then $\hbar, c$ as pure SI conversion.

Already computed.

- **Masses:** S on $\mathbb{C}^3$, eigenvalues² = masses, ratios from $(\sqrt{2}, 2/9)$ (Level 2).
- **Koide:** $Q = (1 + r^2/2)/3$ from the eigenvalue structure, $= 2/3$ at $r = \sqrt{2}$ (Level 2).
- **Process / non-Markov:** the time-extended operator on $L^2(T^4) \otimes (\text{qubit})$, CP-divisibility witness (Level 1).

How further quantities are computed (programme). The same recipe, new operators – the method is fixed, the results are still open:

- **Mixing angles (PMNS/CKM-like).** If two sectors (say charged leptons and neutrinos) are each Z_3 -circulants, but in fibre bases *rotated* against one another, then the mixing matrix is the overlap $\langle f_k^{(A)} | f_l^{(B)} \rangle$ of their eigenvectors. The mixing angles encode the *rotation angle* between the two Z_3 structures – a single geometric number per sector pair, computed just like $2/9$.
- **Further sectors (quarks).** The same scheme: one Z_3 -circulant per charge sector with its own structural angle; the open question is the same – which rationals does the topology force.
- **Dynamics and correlations.** Holonomy and entanglement observables from the base connection Laplacian (Doc. 230), already demonstrated in the witness scripts.

What "computing" means here

Every quantity is an eigenvalue, an eigenvalue relation, or an eigenbasis overlap on $\mathcal{H}$, fixed by structural rationals plus the single scale $\tilde{T}$. That is the sense in which the Hilbert-space mapping "computes further things" – not through new fits, but through new operators following the same recipe. The quark sectors are computed as an operator spectrum (next section) – faithfully, but without pure numbers; the mixing angles remain open. What is fixed is the method.

.7 The three sectors explicitly: faithful translation, lepton-specific numbers

Because the Hilbert-space mapping is a *faithful translation* (the $T^4 \leftrightarrow$ Hilbert bijection, Doc. 270, type III), it must also carry the masses FFGFT obtains via the ξ -route ($m = r \xi^p v$, Doc. 006) – leptons *and* quarks. Here that is carried out: each charge sector is exactly the spectrum s^2 of a Hermitian Z_3 -circulant.

Sector	M	r	Koide Q	max. rel. error
Leptons (e, μ, τ)	17.72	$\sqrt{2} = 1.4142$	0.6667	$2 \cdot 10^{-14}$
Up (u, c, t)	150.7	1.757	0.848	$7 \cdot 10^{-14}$
Down (d, s, b)	25.73	1.546	0.732	$2 \cdot 10^{-14}$

In every sector s^2 reproduces the masses to machine precision (script `ffgft_z3_alle_sektoren.py`). The Hilbert-space calculation is thus *universal* – the quarks are included.

A faithful translation is no gain by itself

This universality is *automatic*: the circle map $\sqrt{m_j} = M(1 + r \cos(\theta + \frac{2\pi j}{3}))$ is a *bijection* (three masses $\leftrightarrow M, r, \theta$). Any triplet can be written into it exactly – it *must*, otherwise it would not be a translation. "It works in Hilbert space" is therefore not a prediction but the meaning of faithfulness (cf. the ξ^N/φ^N triviality, Doc. 190 P35).

The genuine content lies in whether the operator parameters are *pure numbers*:

- **Leptons:** $r = \sqrt{2}$ – which is Koide $Q = 2/3$, the $Z_3 + \sqrt{2}$ structural condition (Doc. 258/259) – and $\theta = 2/9$ (a clean rational). Two pure numbers (see above).
- **Quarks:** $r_{\text{up}} \approx 1.76$, $r_{\text{down}} \approx 1.55$ – *not* $\sqrt{2}$; the angles are not clean rationals. The operator carries the masses faithfully, but with *un-special* parameters.

This matches the ξ -route point for point (Doc. 006): there the lepton ratios carry real structure (Koide), while the quark (r, p) are per-particle classification. A faithful translation *must* agree on *where* the structure is – and it does.

Consequence for a universal Z_3 angle

The attempt to impose the lepton condition $r = \sqrt{2}$ (i.e. $k = 1$, Koide $2/3$) on the quark sectors is *falsified* by the faithful translation itself: the up sector needs $r \approx 1.76$, the down sector $r \approx 1.55$. There is no room for $\sqrt{2}$ – and hence none for a single universal $k = 1$ Z_3 angle across sectors.

.8 The same scale carries the couplings too: α and (briefly) G

The Z_3 operator of the lepton sector fixes not only the masses but also a *scale* M_T and the dimensionless eigenvalues μ_e, μ_μ, μ_τ . The fine-structure constant uses exactly these quantities – it is therefore not a separate input.

α as an eigenvalue relation on $\mathcal{H}$

The very recipe – every quantity an eigenvalue or eigenvalue relation on $\mathcal{H}$ – yields α too, from the *same* operator that gives the masses. The eigenvalues of S are $\lambda_j = \sqrt{m_j}$; the fine-structure constant is the product of the two smaller ones, scaled by ξ :

$$\alpha = \xi \left(\frac{\lambda_e \lambda_\mu}{\text{MeV}} \right)^2, \quad \lambda_e \lambda_\mu = \sqrt{m_e m_\mu} = E_0. \quad (7)$$

This is the ξ -route ($\alpha = \xi(E_0/\text{MeV})^2$, Doc. 011), but as a *Hilbert-space variant*: E_0 is not a separate input but an eigenvalue relation of the same operator built from $\sqrt{2}$ and $2/9$. Thus α and the lepton masses follow from *one* object and *one* scale. Numerically: the operator/geometric value gives $1/\alpha = 138.9$ (bare $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV); with the fractal correction $E_0 = 7.348/\sqrt{K_{\text{frak}}} = 7.398$ MeV, $1/\alpha = 137.04$ follows (R72) – the 1.35 % is, up to 0.017 %, the factor $K_{\text{frak}} = 1 - 100\xi$. (Corrected on 30 Sept. 2026: previously “the corpus tunes $E_0 = 7.398$ MeV to $1/\alpha = 137.04$ – the final near-one-percent sits in the precise value of E_0 (radiative-correction level)” – 7.398 is not tuned but $7.348/\sqrt{K_{\text{frak}}}$; the gap is 1.35 %, and $K_{\text{frak}} \cdot 138.91 = 137.06$.) Script `ffgft_kopplungen_hilbertraum.py`.

G – structurally through ξ , involved only in the SI conversion

The gravitational coupling enters *likewise* through ξ , via the T0 fundamental relation

$$\xi = 2\sqrt{Gm} \implies G = \frac{\xi^2}{4m} \quad (8)$$

(Doc. 006, calc-o). The structural finding is the same as for α : the coupling is *not independent* but fixed by ξ . The Hilbert-space mapping supplies the scale here: m is the operator scale $M_T = M^2 = 1/\tilde{T}$ of the lepton sector. Unlike the dimensionless α , however, G carries dimensions; its SI realisation requires the full conversion through $\ell_P, c, \hbar$ and is therefore considerably more involved – it is carried out in calc-o and yields G_{SI} at the per-mille level. What is involved is the *SI conversion*, not the structural origin.

Masses and couplings from one object

α and the lepton masses follow from the *same* Z_3 operator and its scale; the gravitational coupling is fixed by the *same* ξ . The pure-number content sits in ξ and the operator structure; the SI numerical values – G especially – are conversions, not additional inputs.

.9 The QM translation and what its extension changes

The translation of the QM *operations* has its own document (Doc. 230/231/232): gauge freedom = unitary equivalence, entanglement = holonomy around cycles, Bell correlations = topological connections on T^4 (Doc. 230). That translation does *not* contain mass and

time – it maps the operations only. The natural question: does anything in the QM results change when it is extended – by the $\mathbb{C}^3$ mass fibre (Level 2) and time as process (Level 1)? Computed (ffgft_qm_uebersetzung_erweiterung.py):

Extension	QM result	change
mass ($\mathbb{C}^3$ fibre)	CHSH/Bell on the base	$4 \cdot 10^{-16}$ (none)
time (process)	coherence back-flow	$0 \rightarrow 0.53$
time (process)	min. Choi eig	$+0.0003 \rightarrow -0.0056$

Mass acts on a *separate* tensor factor. A QM correlation (CHSH on two base qubits, $= 2\sqrt{2}$, Tsirelson) is identical to within $4 \cdot 10^{-16}$ with and without the $\mathbb{C}^3$ mass fibre attached. The mass extension only enlarges the space and supplies the masses; *no* static QM result changes.

Time changes the *dynamical* quantities. The QM bridge treats the reduced dynamics as time-local (unitary/Markovian); the extended translation carries the loop history. The same fibre-qubit coherence $c(t)$ is Markovian and CP-divisible with no back-flow, but history-coupled it is non-Markovian: back-flow 0.53, the min. Choi eigenvalue turns negative, $\max |\Delta c(t)| \approx 0.78$. Single-time quantities remain; sequential/dynamical ones acquire the non-Markovian correction.

Slightly incomplete – only in scope

The QM translation is not wrong but *incomplete in scope*: no mass, time-local. Extended, it leaves every *static* QM result unchanged (mass factorises) and adds exactly *one* new effect – non-Markovian memory in the dynamics – which is itself still within quantum theory (a traced-out unitary bath reproduces it, Level 1). The expectation that the extension “has an influence on the results” therefore holds, but precisely located: on *time*, not mass, and only on *dynamical* quantities.

Test duration decides. The difference lives entirely in time and has two scales: a *fast* decoherence time (coherence below $1/e$ at $t \approx 0.4$, nearly gone at $t \approx 1.4$) and a *late* memory/recurrence time (first revival at $t \approx 2.8$, about $6\times$ later). Hence three regimes: (i) very short test ($t \ll 0.4$) – coherence ≈ 1 , QM-only and extended translation identical, the QM result exact; (ii) into the decoherence – both fall essentially together, the Markovian description suffices; (iii) only from the recurrence time on do the curves separate (Markovian ~ 0.53 vs non-Markovian ~ 0.22). Since a single operation typically ends *before* decoherence, the non-Markovian term is *operationally invisible* to fast operations; it becomes measurable only once the test duration reaches the T^4 memory time – in deliberately long, coherence-preserving, sequential experiments.

Does it scale with many qubits or with mass? No – least of all there. Massive systems couple more strongly to the bath and therefore decohere *faster*: at four-fold coupling the decoherence time drops from $t \approx 0.44$ to 0.11 , and the absolute coherence at the first revival falls from 0.22 to $\sim 10^{-5}$ – the non-Markovian term is then *buried*. Mass thus makes it less visible, not more; what remains is faster *ordinary* (Markovian) decoherence. With many thousands of qubits it is not the memory term that grows but ordinary decoherence and error accumulation (gate errors, crosstalk) – Markovian and addressed by error correction; whether revivals survive at all depends on the bath *structure* (incommensurate frequencies push the Poincaré recurrence into the practically unreachable), not on the qubit count. The one legitimate caveat is *correlated* (non-Markovian) noise across gates – relevant only if the bath memory time approaches the sequence duration, which fast gates and fast-decohering (heavy) systems precisely prevent.

.10 Compact or recursive? When the compact computation suffices

One question decides how far the compact Hilbert-space computation reaches: does the compact (cyclic, finite, one-shot) variant *always* suffice – so that decompactification is only the final step to SI – or are there quantities computable *only recursively*? The decisive criterion is **linearity**.

Linear $\Rightarrow$ compact suffices

If the dynamics is generated by a *fixed linear* operator on $\mathcal{H}$ with a discrete (compact) spectrum, the closed spectral representation always exists:

$$\langle A(t) \rangle = \sum_{k,l} c_{kl} e^{i(\omega_k - \omega_l)t}. \quad (9)$$

One diagonalisation suffices. The memory kernel $K(t)$, which looks *recursive* in the time representation (convolution, memory), is merely the Fourier transform of the discrete spectral density $\sum_k g_k^2 \delta(\omega - \omega_k)$. The recursion is thus only *one representation* (time domain); the spectral sum is the equivalent closed one. For a discrete spectrum the closed form always exists – decompactification happens only at the end, at the SI step.

Concrete evidence: the CP-divisibility test (Level 1) obtained the *non-Markovian* result – back-flow, revivals – from exactly such a finite spectral sum, in closed form, *without* recursion. Even “memory” was compact-computable.

When recursion is forced

Recursion becomes unavoidable only if one of the two linearity prerequisites breaks:

1. **Fundamentally continuous spectrum** – then no finite sum. But the continuum appears only at the SI/experienced level; fundamentally it is compact. This case drops out.
2. **Nonlinear / self-referential dynamics** – the generator depends on the *state or the history itself*, not only on time. Then there is no fixed eigenbasis, no diagonalisation, and the recursion is genuine.

The question thereby reduces to a *classification*: for each quantity ask – is the generator a fixed linear operator on $\mathcal{H}$? Yes $\Rightarrow$ compact, one-shot, SI only at the end. Does it depend on the state/trajectory $\Rightarrow$ recursion possibly irreducible.

Mass sector, gauge, entanglement and mixing are linear – for them the compact computation suffices, and HLV's recursions are *avoidable* there. The only candidate for genuine recursion is the *self-referential* level (Doc. 248, §8: the origin thought as a material T^4 process): where the process processes itself, the generator could become state-dependent. To check this – linear or self-referential, quantity by quantity – is a well-defined programme and marks the single place where the open/recursive variant would do real, non-eliminable work.

.11 Claim-state and limits

What is shown – and what is not

1. **A genuine Hilbert-space mapping.** The mass operator is a concrete Hermitian operator on $\mathbb{C}^3$ with eigenvectors (generations) and spectrum (masses) – not a parametrisation.
2. **Koide forward.** $Q = 2/3$ follows exactly from $r = \sqrt{2}$; structure, not a fit.
3. **Two numbers, three masses.** $(\sqrt{2}, 2/9)$ reproduce both ratios to 4–5 digits; one angle covers two observables (a genuine over-determination).
4. **Structural assumptions.** The *form* (Z_3 -circulant) and $r = \sqrt{2}$ are claimed geometric (Doc. 206/258/259).
5. **Empirically exhausted.** $\theta = 2/9$ is confirmed against the e/μ data to $\sim 2 \cdot 10^{-7}$ (seven figures in the angle); the ratios agree to 10^{-5} – $6 \cdot 10^{-5}$, for the τ within the measurement uncertainty. More precise lepton data are not to be expected – the empirical question is answered. The tiny residual ($Q = 0.6666605$ vs. $2/3$) sits at the level of radiative corrections, not a defect of the identification.
6. **The only open question is theoretical:** is $\theta = 2/9$ (like $r = \sqrt{2}$) forced by the T^4/Z_3 topology?
7. **Not beyond quantum theory.** The CP test (Level 1) shows non-Markovianity in the reduced dynamics; a traced-out unitary bath reproduces it.

The sharp, now reachable remainder is thereby precise: not “derive a messy number”, but the single clear question – **is $\theta = 2/9$ forced by the T^4/Z_3 topology, as $2/3$ is?** A single rational, not a ladder fit.

Topology frames, dynamics determines – the reading that was already right. The answer that carries here was at heart always the dynamical one: a mass ratio is *only* dynamically thinkable. The spectrum $\sqrt{m_k}$ is the solution of an eigenvalue problem, and the fixed point $\mathcal{R}(\Phi_*) = \Phi_*$ (Doc. 203) is *both at once*: the topology (Z_3, T^4) fixes the *form*, self-consistency the *value*; the two cannot be separated. A computed finding (`kann_topologie_2_9_erzwingen.py`) *supports* exactly this reading: $\theta = 2/9$ *rad* is not a static flux root of unity – $\theta/2\pi = 1/(9\pi)$ is irrational, and by Lindemann-Weierstrass $e^{i\theta}$ is transcendental. A *purely combinatorial/counting* topology (flux quantization of a finite group) therefore yields only the *spacing* $2\pi/3$ (Z_3 , the denominator; $Q = 2/3$ is forced for *every* θ by $r = \sqrt{2}$ alone), not the offset. But this does *not* pit topology against dynamics: transcendence-in- 2π is here *no prohibition* but the *expected* signature of a dynamically (rather than combinatorially) determined quantity – a counted value would be a root of unity, a dynamically set one typically is not. Properly posed, the open question is therefore: the topology frames (Z_3 spacing, the form), a *dynamical* condition on that stage fixes the offset – and *which* condition yields exactly $\theta = 2/9$ is the hard, still open core. Excluded is only pure counting topology, not topologically framed dynamics (the fixed point itself). (*Aside*: the value $2/9$ also appears in the corpus as a dimensionless mode-locking rotation number, Doc. 286; this is *not* dimensionally the same as the radian angle $\theta - 2\pi \cdot 2/9 = 4\pi/9 \neq 2/9$ –, so the identification remains open.)

Why $2/9$ cannot be had symmetrically (the $\cos 3\theta$ theorem). There is an exact reason why no simple *symmetric* principle yields $\theta = 2/9$ (`einfachstes_prinzip_2_9.py`). The entire θ -dependence of the mass configuration $a_k = 1 + \sqrt{2} \cos(\theta + 2\pi k/3)$ runs *only* through

$\cos 3\theta$: the elementary symmetric functions are $e_1 = 3$ and $e_2 = 3/2$ (both θ -independent), only $e_3 = \det = -\frac{1}{2} + \frac{\sqrt{2}}{2} \cos 3\theta$ carries θ . Hence every Z_3 -invariant quantity extremizes solely at $3\theta = n\pi$, i.e. $\theta = n\pi/3$ – the Z_3 -symmetric points $\{0, \frac{\pi}{3}, \frac{2\pi}{3}, \dots\}$. Concretely, the smallest-denominator, the widest-tongue and the spectral-determinant principles all select $1/3$ or $1/2$, never $2/9$. The reason is a structural theorem, not a computational shortfall: *symmetric functionals cannot have asymmetric values as extrema*, and $\theta = 2/9$ rad lies about 21 % of the way between 0 and $\pi/3$, irreducibly Z_3 -breaking. (Corrected on 1 Oct. 2026: previously “ $2/9$ lies $2/3$ of the way between 0 and $1/3$ ” – the Z_3 -symmetric points lie at $\theta = n\pi/3$ rad; $(2/9)/(\pi/3) = 0.212$. The reading “ $2/3$ between 0 and $1/3$ ” holds only for the rotation number, which is explicitly separated from the radian angle above; the conclusion that $2/9$ is not a symmetric point remains correct.) Thus the simplicity principle is not refuted but shifted: not the *simplest symmetric fraction* ($= 1/3$), but the *minimal symmetry-breaking value* – and the holonomy test (Doc. 286) shows that $2/9$ is exactly this: the only member of the 9-family that splits off as a singlet under the holonomy ($1/9, 4/9$ stay paired). Simplicity sits in the *breaking*, not in the symmetric ground state; what remains precisely open is only the defined symmetry-breaking measure whose minimum yields $2/9$ (rather than $1/9$ or $4/9$).

Test: the breaking reaches $2/9$ but does not fix it. A direct test (symmetriebrechung_2_9.py) couples the second triplet $3'$ of the doubling ($6 = 3 \oplus 3'$, Doc. 285) with strength δ and relative phase χ and looks for the shifted extremum θ^* of the mass configuration. Result: the breaking *reaches* $\theta^* = 2/9$ – so it is not only necessary but sufficient –, but at a strength δ that must be tuned ($\delta \approx 0.52$ at $\chi = 0$, 0.24 at $\chi = \pi/2$) and coincides with no independently fixed value (ξ , $1/\sqrt{2}$, $1/\varphi$, $1/3$). Thus $2/9$ is *reachable but not point-derived* through this functional: the breaking strength remains a free modulus – consistent with the transcendence and the free holonomy phase. The open question is thereby precisely shifted – no longer “why $2/9$ ” but *what fixes the strength of the Z_3 breaking*. (Caveat: this is *one* natural breaking functional; it does not exclude that another fixes δ independently – e.g. the self-consistency of the fixed point with the fixed lepton exponents, Doc. 203.)

The common reason: everything determined is radial, θ is angular. Precisely this fixed-point route was tested (fixpunkt_radial_vs_winkel.py) – and it does not bind θ . The fixed point $\mathcal{R}(\Phi_*) = \Phi_*$ (Doc. 203) with the condition $\xi_0^{p_i-1} K_{\text{frak}} = 1$ and the exponents $(3/2, 1, 2/3)$ does not determine ξ_0 (cf. the notes in Doc. 203); θ does *not* appear in it. (Corrected on 1 Oct. 2026: previously “fixes ξ_0 via $\xi_0^{p_i-1} K_{\text{frak}} = 1$ from the exponents $(3/2, 1, 2/3)$ ” – for $p_\mu = 1$ the condition requires $K_{\text{frak}} = 1$ (actually $74/75$), for $p_e = 3/2$ it gives $\xi_0 = K_{\text{frak}}^{-2} = 1.027$, for $p_\tau = 2/3$ $\xi_0 = K_{\text{frak}}^3 = 0.961$: no common ξ_0 and in no reading $4/30000$. The decoupling of θ and ξ_0 holds *nonetheless*, since θ does not appear in the condition.) The radial sector (ξ_0 , hierarchy) and the angular sector (θ , Q) are decoupled: ξ_0 is θ -free, $Q = 2/3$ is ξ_0 -free. The same pattern runs through the whole investigation: *everything FFGFT determines is radial* – ξ , the exponents, the hierarchy, $Q = 2/3 = r=\sqrt{2}$, the mean $\langle f \rangle$ – and *everything open is angular* – $\theta = 2/9$, the holonomy phase, the harmonic phases of f , the breaking strength δ . Fixed point, duality, recursion and the torus counting apparatus all live in the radial sector without exception; none touches the angle. This is the common structural reason why $\theta = 2/9$ resists every tool. To fix the offset one needs an *angular* principle – a dynamical angle condition – which FFGFT does not yet possess and which, by the transcendence finding, cannot be a counting/topological one.

(Note of 2 Oct. 2026 on notation: here $\langle f \rangle$ is the mean of the fractal correction function in $g = \eta(1 + \xi f)$, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

Not static-geometric either – closing the elimination. There remains the question whether the angular principle is a *fixed geometric angle* of the shared icosahedral stage on which FFGFT's 3-fold (C_3) and HLV's 5-fold (C_5) coexist ($C_3 < A_5$, Doc. 283). Tested (ikosaeder_winkel_2_9.py, strict, hit threshold 0.05°): the icosahedral axis angles are 20.9°, 31.7°, 36°, 37.4°, 41.8°, 54.7°, 58.3°, 60°, 63.4°, 69.1°, 70.5°, 72°, 79.2°, 90°. $\theta = 2/9 \text{ rad} = 12.73^\circ$ matches *none* (nearest 20.9°, distance 8.2°), nor do 2θ or $\theta/2$. Control: $\arccos(1/\sqrt{5}) = 63.43^\circ$ matches the C_5 – C_5 angle *exactly* – the method is correct, so the negative is real. (Koide's 45° is not an axis angle either: FFGFT's angular structure is not the icosahedral geometry.) This closes the elimination: $\theta = 2/9$ is *not* symmetric ($\cos 3\theta$ theorem), *not* counting-topological (transcendence), *not* radial (fixed point), *not* static-geometric (this computation). Exactly *one* category remains – the *dynamical skew-adjoint phase* R (Doc. 283), an accumulated, Berry-like angle. FFGFT's own recursion $\mathcal{R}$ (Doc. 203) is purely real and possesses no such generator; only the HLV side carries it ($R = \text{BD17A}$). The static-radial FFGFT side is thereby *demonstrably exhausted* for θ ; the angular principle is structurally HLV's dynamical R , with a precise target: an accumulated angular phase yielding $\theta = 2/9$ – transcendental in 2π , Z_3 -breaking, the singlet of the 9-family.

.12 Conclusion

Conclusion

FFGFT's Hilbert-space translation reaches further than previously shown. Level 0 maps the QM operations cleanly (shared with HLV). Level 1 brings time as process (CP-divisibility, non-Markovianity). Level 2 brings time as geometry and hence the *masses*: the mass operator is a Hermitian Z_3 -circulant on the $\mathbb{C}^3$ fibre whose spectrum yields the lepton masses from two pure numbers $\sqrt{2}$ and $2/9$ – Koide exact, both ratios to 4–5 digits, without the old fitted coefficient. $2/9 = 2/3^2$ is a structural number on the footing of ξ , a foundational condition of the fractal space. The whole lepton problem is thereby reduced to a single question – and this document's investigation has made it *angular*: not *whether* T^4/Z_3 forces the angle (no counting topology can, $2/9$ being transcendental in 2π), but which *dynamical* angle principle yields it. The elimination shows: everything determined is radial (ξ , hierarchy, $Q = 2/3 = \sqrt{2}$), only the offset $2/9$ is open – structurally HLV's skew-adjoint dynamical phase R .

Closing remark: translation is not justification. Simplified, the result of these extensions fits in one sentence: *a faithful translation into Hilbert-space notation is possible for all masses and all constants – but precisely because it is faithful, it provides no justification.* Faithfulness is mathematically a bijection: three masses correspond invertibly and uniquely to the triple (M, r, θ) , α is an eigenvalue relation of the same operator, G follows through the same scale $1/\tilde{T}$. A bijection, however, carries *everything* over and explains *nothing*; it is a change of coordinates, not a derivation. That a quantity “can be computed in Hilbert space” is therefore never a finding in its own right – it is the meaning of faithfulness.

The justification comes from the original T^4 torus representation, which is generatively prior (Doc. 206/270): the Hilbert space is the *image*, the torus the *original*. Only there are $\sqrt{2}$ (the Koide condition $Q = 2/3$), the angle $2/9$ and the primitive ξ *geometric resp. topological facts*

rather than read-off parameters. The clean division of labour is thus: the Hilbert space *computes and represents* (universally – leptons, quarks, α , G); the T^4 *justifies and derives*. Even where the translation yields *no* pure numbers – for the quarks – this is not a defect of the translation but an expression of the fact that the torus geometry forces none there. The open question of the whole construction is therefore a single one, and it must be posed on the torus side: whether T^4/Z_3 forces the pure numbers $\sqrt{2}$, $2/9$ and ξ .

Accompanying scripts (numpy). `ffgft_t4_cp_divisibility.py` (Level 1, CP-divisibility), `ffgft_t4_mass_operator.py` (Z_3 -circle analysis, KK comparison), `ffgft_mass_operator_C3.py` (the mass operator on $\mathbb{C}^3$), `ffgft_z3_alle_sektoren.py` (all three sectors as an operator spectrum), `ffgft_kopplungen_hilbertraum.py` (α and G via the Hilbert-space mapping), `ffgft_qm_uebersetzung_erweiterung.py` (mass/time extension of the QM translation).